



Smart spraying technologies for precision weed management: A review

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ABSTRACT

In this study, the previous work, present status, benefits, and limitations of the state-of-the-art technologies used in smart spraying technologies in precision weed management are reviewed. A total of 116 articles were identified from Google Scholar and Scopus to study the research work in the field of smart sprayers and precision weed management. The articles were examined based on the relevance, research focus, novelties, measured parameters, used technologies, and field of applications. Smart sprayers based on machine vision (MV) and artificial intelligence (AI) are keys to improving crop productivity and meeting the food demands of the future by reducing the yield losses due to weeds and working towards a sustainable future in agriculture. Many papers published in recent years have focused more on the machine vision, weed detection, and classification aspects of the weeding robot. Very few studies have attempted to discuss the components of a smart weeding machine, non-chemical-based weeders, the components of spraying systems, their controls, underlying fluid mechanics, and the field trials of these weeding robots. This article reviews conventional weeding techniques, machine-vision-based weeding robots, and spraying systems proposed or constructed in the last twenty-five years. Key technologies such as non-chemical-based weeding machines, image preprocessing, feature extraction, and weed detection based on machine learning (ML) and deep learning (DL) for smart sprayers are discussed. The fundamental components of a smart spraying system are also discussed, and previous works are compared to highlight the key components, the spraying accuracy, and the major advantages and disadvantages. The fluid mechanics of the spraying system and its associated controls involved are also presented. There are still many bottlenecks in weed detection systems and smart spraying systems. The results of the systematic review provide an understanding of the progress made in the field of robotic weed detection, herbicide and non-herbicide-based weed management, the use of machine vision, and the limitations of the current spraying systems.

1. Introduction

Yield losses in crops because of the presence of weeds which compete with plants for nutrients, water, space, and sunlight have been documented for different plant and weed types all over the world. A study conducted in the Philippines [1] estimated the loss in yield to be between 44 and 96 % for different rice cultures when the weeds were left unchecked. Sinden et al. [2] calculated the total costs of weeds to Australian agriculture which included losses in the crop, livestock, and horticulture industry. They estimated the total mean loss to be between AU\$3583 million to AU\$3737 million (\$2.1 billion to \$2.2 billion USD) per annum based on the data collected between 1997 and 2002. This includes the sum of total financing costs of on-farm weed control and the costs because of lost production. This estimate is a combined value for

multiple industries including grain, dairy cattle, beef cattle, sheep, cotton, rice, etc.

Weed control is especially important in vegetable crops for numerous reasons. Weeds negatively affect the yield of vegetables by reducing their quality and market value [3]. Plants can detect the presence of competing crops (in this case the weeds) nearby and try to use the available resources to outgrow the competing plant. This could result in lower-quality vegetable production as the available nutrients and resources are diverted to compete for external resources (e.g., water and sunlight) [4]. William and Warren [5] conducted studies in Brazil to determine the crop losses due to purple nutsedge competition. The presence and the resulting competition of purple nutsedge for water and light resulted in crop losses of 89 % for garlic, 62 % for okra, 53 % for tomato, and 43 % for cucumber. Despite the widespread use of

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herbicides in the USA, certain vegetables recorded a loss in value between 8 and 13 % due to weeds [6].

Weeds can act as a host for insects that attack vegetable crops [7]. This creates added problems related to insect control and prevention of vegetable crops getting damaged by insects (e.g., whiteflies in tomato plants; vegetable crops such as sugar beet, radish, lettuce, broccoli, spinach, etc. may be affected by the green aphids which are hosted by wild mustards; [8]). Weeds can also act as a host for disease-causing pathogens that can damage vegetable crops. Palmer amaranth, a common weed found in Florida can host the Tobacco rattle virus which causes yellow streaks on plants such as potatoes and sugar beet [9]. An area with a low or non-existent weed population can see rapid growth of weeds without focused weed management efforts and their weed impacts could be passed on through multiple generations (Young & Pierce, 2014).

Thus, weed control becomes important for maintaining the quality of food, especially vegetable crops, reducing the impact on yield, and preventing associated financial losses.

2. Methodology

This systematic review focuses on the ground-based smart spraying technology of the past 25 years. Over this period, we have seen numerous developments in the field of image processing and machine vision (MV) methods, machine learning (ML) algorithms, deep learning (DL) algorithms, and robotic spraying systems. The machine vision system and spraying system are given more focus in this review. The keywords “weed management”, “weed detection and spraying”, and “robotic weeder” were used to search and select related literature for machine vision and spraying systems by applying a keyword-based search in Google Scholar and Scopus for journal and conference papers. Based on a detailed analysis of these works, a review of weed sprayer systems is highlighted. Since 2015 there has been a huge growth in the number of papers published in this area. This has been driven by the advances in research related to DL algorithms, improved processing power of machines, availability of low-cost cameras and graphical processing units (GPUs) and increased general adoption of DL and ML techniques to agricultural applications. The search results breakdown for Scopus is given in Fig. 1.

3. Weed control without using herbicides

Weed control is carried out either between crop rows, within crop rows, or a combination of both. Inter-row weeding using cultivators involves destroying the weeds by partially or completely burying the weeds, uprooting them from the soil, or breaking off the roots from the soil. The major limitation of the cultivators is that cultivation must be performed at the initial growth stage of the crops to avoid damage to the crops through machine-plant contact. Intra-row systems require machinery with special mechanisms and add-on tools to perform weed control within the crop row and close to the crops while avoiding any contact with them. The machines could either have a sideways movement of the weeder or have weeding equipment that performs weed control through sideways movement [10].

3.1. Mechanical systems

Mechanical weeding involves the use of tools, actuators, and mechanical components that control weed growth or eliminate them through direct physical contact with the weeds and/or the soil in a single or repetitive motion. The number of passes and the intensity of each pass depends on the density of the weeds, the density of the plants, the distance between the rows, and the stage of weed growth [11,12]. Harrowing is an old technique that has been used since ancient times to uproot small weeds in the early stages and was the most popular choice in Europe in the late 1980s as an alternative non-herbicide-based weed control method [13]. These harrows can have spiky tips with different tine angles, shapes, and depth profiles but often require repeated passes to achieve sufficient control of weeds during the entire growing season. Van Der Weide et al. [14] also report the increased popularity of finger weeders and torsion weeders in European weed control practices in the early to mid-2000s because of their much better performance as compared to harrows (Fig. 2). Brush weeding, which uses a mechanism similar to finger weeders but with brushes on each side of the row driven by hydraulic motors to create an uprooting of the intra-row weeds, is another mechanical weeding technique [15]. MSR Technologies [16], a Danish company, developed a mechanical weeder that targets weeds growing on ridges and on side slopes. However, it is limited to potatoes and might not work well with other crops that have intra-row weeds. Low-cost ultrasonic sensors were used in an intra-row mechanical weed control prototype with a rotating pinch-roller weeding mechanism for weed control in vegetable crops [17]. The weeder pulled weeds between

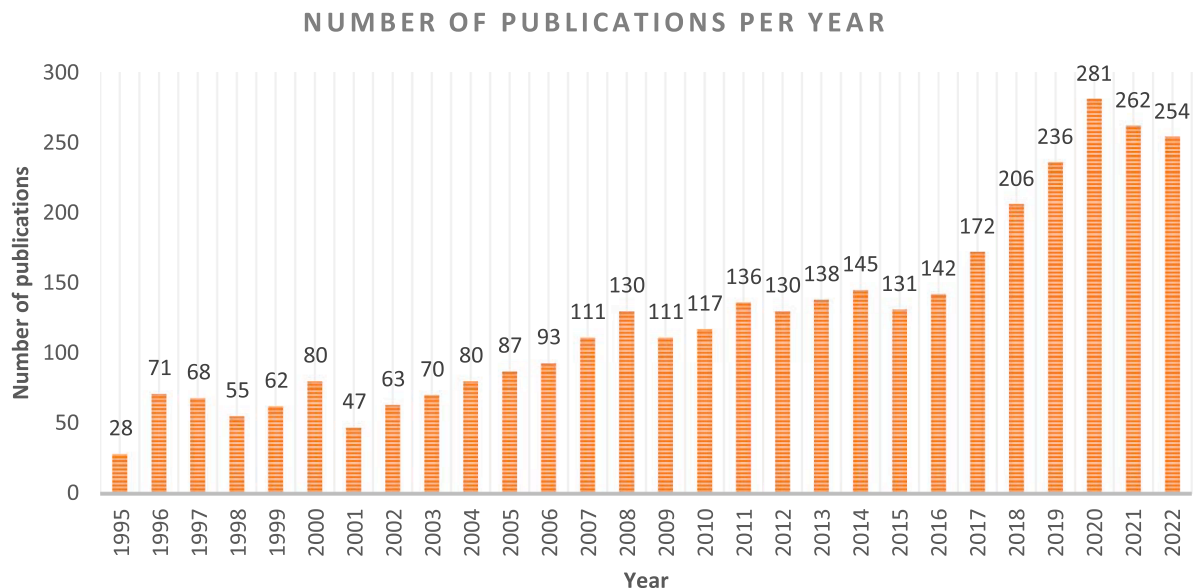


Fig. 1. Number of publications per year as per Scopus search results for the keywords “weed management”, “weed detection and spraying”, and “robotic weeder”.

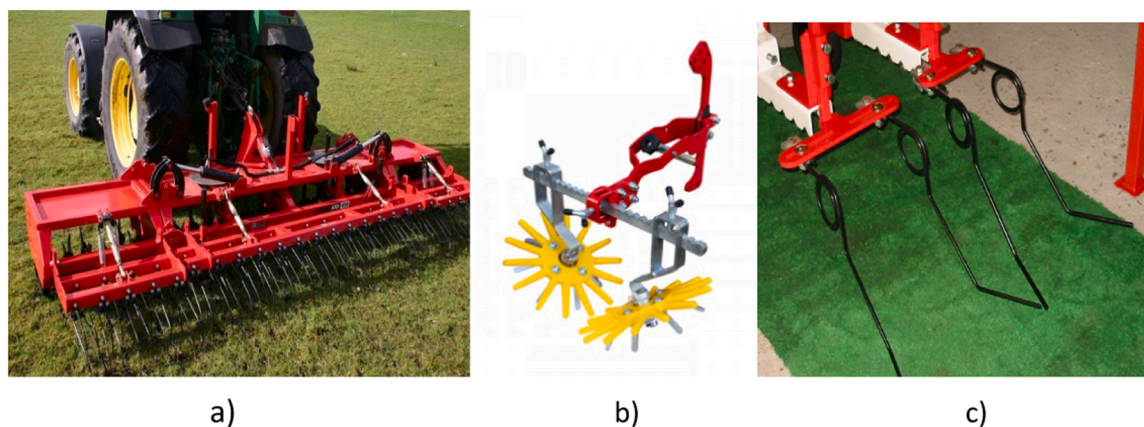


Fig. 2. (a) Harrowing equipment attached to a tractor (<https://toptillers.com/harrowing-guide/>) (b) Finger weeder (terrateg.com) (c) Torsion weeder (<https://www.frato.nl/UK/torsiewieder-UK.htm>).

8 and 13 cm in height and the best performance was seen on Southern crabgrass weed (34% weed control efficiency).

The issue with most of the mechanical weeders mentioned above is the lack of a mechanism to distinguish between weeds and plants. They work mostly on large inter-row spacing of the crops and fail when the spacing is narrow or when there are more weeds in between the crops within a row [14].

3.2. Laser treatment

Laser treatment can also kill the weeds by focusing a high-energy laser beam which either cuts the weed, burns, or just releases enough energy to stunt their growth. The effectiveness of laser weed treatment depends on the wavelength, laser power, exposure time, and spot size [18]. Two continuous wave diode lasers were used on common chickweed, scentless mayweed and oilseed rape grown in potted plants and the exposure of apical meristem of these species to laser resulted in physical control of weeds with a 5 W, 532 nm and a 90 W, 810 nm laser. However, some species (*B. napus*) are less susceptible to laser treatment and require high doses for higher weed control. Shah and Lee [19] used 16 lasers distributed in two rows of eight each mounted on a four-wheeled robot cart. The lasers were triggered based on detections of green color on a soil background after Hue Saturation Value (HSV) transformation. The laser system showed an accuracy of 63.6 % in eliminating weeds. Xiong et al. [20] used two 5 V class-1 laser pointers and performed static and dynamic weeding method evaluation. The weeding experiments were performed in indoor environments and had a high average hit rate of 97 %. However, this would fail in complex cases where the combination of color segmentation and the morphological operation are not enough to detect weeds from the plants. A low-cost laser system was used to thermally damage the stems of couch grass (*E. repens*) which were mixed with tomato plants [21]. Three laser types (1 W, 3 W, and 5 W) were tested to determine the power required to cut weeds with a diameter greater than 2 mm. The 1 W laser proved ineffective as it could not provide enough power to cut the stem. The 5 W laser beam was the most effective in cutting the weeds, however, the risk of damaging the crop was also higher because of the splitting of the laser beam. It was also recommended to use the laser system in the early weed stages, especially when the weeds have developed 2–4 leaves.

3.3. Thermal treatment

Exposing the weeds to high temperatures either through hot water or steam is another non-chemical-based method of weed control. A hot water treatment system developed in New Zealand, called the Waipuna system (Waipuna, Auckland, New Zealand) used a combination of hot

water (95°C) and a foam (a mixture of coconut oil and maize oil) to kill weeds [22]. The hot foam system provides an insulating effect and keeps the hot water in contact with the weeds for a longer time thereby increasing their effect [23]. However, this system was not made available in other parts of the world and the hot water system had a few disadvantages. The energy associated with creating hot water and the requirement of carrying large volumes of water makes it impractical for large-scale applications, especially in regions where water availability is an issue. This method is environmentally friendly and can still be used in smaller farms in the initial stages of weed development [24].

The steam-based weeding equipment release vaporized water which has a considerably lower water use per unit area and a higher heat transfer coefficient as compared to hot water-based weeding machines. But the increased energy losses in steam equipment make it a far less viable alternative [23]. Abdulridha et al. [25] evaluated a weed steamer with a high (53 L/h) and a low flow rate (15 L/h) and established that the high flow rate killed 93–97 % of weeds within three days of application. Superheated steam offers better coverage and penetration as compared to hot water and even exposure for 2–4 s can kill a variety of weeds [24]. Kolberg and Wiles [26] compared two different application rates of steam (3200 kg/h and 1600 kg/ha) on cropland weeds and discovered that the higher application rate reduced weed biomass nine weeks after application. They also concluded that the effectiveness of weed control by steam depends on the application rate, type of weed, and growth stage. Flame weeders [27,28] and infrared weeders [29] have also been implemented in the past but they either pose a fire risk (flame weeders) or have lower efficiency and suffer from shading interference in dense vegetation (infrared weeders) [23].

3.4. Robotic systems

The limitations of hand weeding, mechanical weeding, and other manual methods that use the weeding techniques discussed above are that they require either a larger gap between the rows and target only the inter-row weeds, or are slow, inefficient, and suitable only for small-scale farms because of human dependency. This necessitates the need for robotic systems with the capability to correctly identify and target both inter-row and intra-row weeds. These systems could either use chemical weeding, mechanical weeding, thermal weeding, or a combination of these. Åstrand and Baerveldt [30] developed a prototype robot with a mechanical weeding tool for in-row weeding guided by a color vision system to identify weeds and a gray-level vision system for row guidance. 'RemoteFarming. 1', a collaborative research project between the University of Applied Sciences Osnabrueck (Germany), Robert Bosch GmbH, and Amazonen-Werke (Germany), is a multipurpose field robot platform. It tackles the problem of robotic weed control (intra-row

weeding) in organic carrot farming using machine vision and a mechanical weeding tool with a sharpened head that damages the weed and pushes it deep into the soil to prevent further growth [31]. The first stage of development required a remote human worker who marked the weed positions which was then transferred to the robot to activate the mechanical tool. The second stage also depended on human input to confirm the weed positions based on the colored overlay pictures. This dependence on human input at different stages would slow down the real-time application and increase human errors affecting the detection accuracy. A robot platform called the Intelligent Autonomous Weeder (IAW) was developed for autonomous weed control in a maize field using mechanical actuators [32]. It performed autonomous weeding on 18 parallel tramlines each of 40 m in length while moving at a speed of 0.5 m/s. Raja et al. [33] developed a mechanical, real-time, fully automated weed knife control system to kill intra-row weeds that were close to the lettuce and tomato plants. However, it had a low weed detection accuracy of 83 % and used a crop signaling technique using fluorescent straws which had to be placed along with the seedlings in the greenhouse to pick up the fluorescent color. This technique might not be widely adopted by farmers because of the need to use water-based compounds on all of the plant seeds for better detection.

Many commercial robotic weeding machines have also been developed over the past 10 years. Garford Farm Machinery (Peterborough, UK) developed the *Robocrop In-row* weeder which is a mechanical weeder for transplanted crops like cabbage and lettuce where the mechanical arms move in a semi-circular motion around the plants to remove weeds as close as 8 mm to the crops [34]. Naïo technologies (Escalquens, France) have developed a large-scale weeding robot (Dino) for lettuce, onions, carrots, etc., and a vineyard weeding robot (Ted). They use different mechanical weeding mechanisms such as finger hoes,

mechanical inter-vine tools, serrated discs, and disk lump breakers. Bosch's Deepfield Robotics team developed an autonomous robot called BoniRob (built on the RemoteFarming. 1 platform; [31]), that can mechanically smash weeds with a ramming rod in carrot cultivation trials. It uses decision tree learning to learn important distinguishing parameters such as shape, leaf color, and leaf size to differentiate between plants and weeds. The mechanical weeding mechanism of BoniRob (Fig. 3(c)) is claimed to be 90 % effective in eliminating weeds [35]. Seattle-based company Carbon Robotics (Fig. 3(b)) developed a laser weeder with sub-millimeter accuracy that covers 2 acres/h at a speed of 1.6 km/h [36]. However, it needs a Lidar for safety purposes to scan farm workers and systems to avoid the 150 W CO₂ lasers becoming a safety issue. Table 1 presents a list of high-performing non-herbicide-based weeding systems, the method of weeding used and their references.

Out of the existing mechanisms, robotic systems often provide better performance in weeding because of advanced detection and weed elimination capabilities. Thermal weeding systems with hot water and steam have limitations and present an added safety risk. Laser weeding mechanisms would need high-intensity lasers which add on additional costs. Mechanical weeding has an issue related to limited workability, i. e., it needs favorable soil conditions for it to work well. Because of the mechanism of punching systems that are employed in some mechanical weeders, the soil structure must normally be fine and easy to work, and lumpy or stony soils might not be favorable [14].

4. Weed control using herbicides

Precision spraying is the most widely used implementation because of its targeted application and has higher reliability as compared to



(a)



(b)



(c)



(d)

Fig. 3. Some commercial and research non-herbicide-based robots (a) ARA model from Ecorobotix (b) LaserWeeder 2021 from Carbon Robotics (c) BoniRob from RemoteFarming. 1 project (d) Agbot II from the University of Queensland.

Table 1

Examples of some high-performing non-herbicide-based weeding systems/robots (research and commercial).

Weeding method	Example of system/robot	Crop used on	Performance	Refs.
Mechanical Weeding (non-robotic)	Optiweeder	Potatoes	Only for inter-row. Accurate weeding until 1 cm from the crop.	[16]
Laser weeding	Carbon's Laser Weeder	Carrots, cauliflower, broccoli	Sub-millimeter accuracy; kills up to 200,000 weeds/h	[36]
Thermal Weeding	Kolberg and Wiles' steam weeding	Tested on croplands with various crops.	Greater than 90 % control of common lambsquarters and redroot pigweed	[26]
Robotic Weeding	Bonirob (Mechanical Weeding mechanism)	Carrot, sugar beet	Achieved a weed control rate of 93.86 %.	[31, 37]

previous methods [38]. The major issue associated with it is the detrimental effect of the applied chemicals on the environment and therefore most weed control methods are trying to optimize the amount of spraying to achieve an overall reduction in spraying.

4.1. Conventional herbicide treatments

The conventional method of spraying herbicide involves blanket spraying, i.e., spraying the same quantity of herbicide throughout the field irrespective of the weed distribution. This is the most popular conventional method of weed management. The herbicide application in conventional spraying could be either soil application or foliar application [39]. Soil applications include soil surface applications where the weed germination in the top layer is targeted, the sub-surface application where herbicides are applied to the lower layer of soil by injecting herbicides at different points either manually or using a machine, and through band application where a narrow band application over the crop row is performed by boom sprayers. Foliar applications could include blanket application over the entire leaf area or direct application in between crop rows. Both foliar applications could be done using a boom sprayer.

Conventional spraying requires substantially less initial investment in equipment or machinery than precision “see and spray” systems. Blanket and broadcast spraying also take less time as compared to some site-specific weed management methods. The major disadvantage is the excessive use of herbicides which results in higher daily costs, higher environmental pollution, and crop damage. Dependence on manual labor also incurs uncertainty due to the unavailability of manual labor and additional costs for every weeding session. Availability of manpower, human error associated with intra-row spraying, and lack of control of spray drift are major challenges associated with conventional treatments.

4.2. Precision and smart technologies

Precision agriculture uses automation and modern technological advancements (machine vision, high-resolution cameras, processing power, robotics, and nozzle technologies) to optimize the use of resources with precise variable rate application [40]. Precision and smart technologies use targeted spraying on the weeds while avoiding the crops, as opposed to the conventional spraying methods which are either blanket sprayers for inter-row spraying or need human interference for intra-row applications.

Lee et al. [41] developed a precision herbicide spraying prototype robot for tomato plants. It used a machine vision system based on image processing algorithms that correctly identified 73.1 % of tomatoes and

68.85 % of weeds in real-time. Åstrand and Baerveldt [30] developed a prototype robot with a mechanical weeding tool for in-row weeding guided by a color vision system to identify weeds and a gray-level vision system for row guidance. Commercial precision spraying systems also exist. Garford Farm Machinery (Peterborough, UK) developed the robotic spot sprayer with specialty nozzles to treat weeds in potato, carrot, parsnips, and onion farms. Blue River Technologies (Sunnyvale, California; acquired by John Deere in 2017) developed a smart see and spray technology for weed identification and spraying. They use machine learning and computer vision technology to achieve weed versus crop differentiation and pair it with a tractor or agricultural farm machine to perform weeding at 19.3 km/h [42]. Ecorobotix (Switzerland) developed an ultra-high-precision (6 cm x 6 cm) smart sprayer (ARA model, Fig. 3(a)) which works for a driving speed of up to 7 km/h and achieves coverage of 4 ha/h. A rack of 156 nozzles were used, that were closer to the ground and were covered on each side to protect against wind and hence reduce spray drift [43].

4.2.1. General components of a precision weed sprayer

The fundamental components of a precision weed spraying system are [44]:

- (1) A weed sensing system (vision system): This includes a camera and/or sensor system that captures images, identifies the location, and type of weeds and plants, localizes them, and relays the necessary information to the processing system which could be either based on image processing, machine learning, deep learning or a combination of one or more of the three.
- (2) A weed management model that uses information from the decision-making algorithms, crop-weed population dynamics, and optimizes the treatments (type of herbicides, volume of herbicides used) based on the type, density, and location (proximity to crop) of the weeds.
- (3) Spraying system which receives commands from the sensing system and decision-making weed model algorithms and uses it to trigger specific nozzles on either a boom or non-boom section.
- (4) Driving platform: The weed sprayer and sensing system are attached either to a robot (autonomous, semi-autonomous) or to a farm vehicle like a tractor. The power to drive the systems mentioned in parts a to c is derived from the driving platform either directly from the vehicle or batteries mounted on the platform.

4.2.2. Vision system

Weed detection using computer vision technology can be broadly categorized into two methods. The first method uses traditional image processing techniques and the second uses deep learning algorithms.

4.2.2.1. Detection using image processing methods. This section reviews research work that involves weed detection using image processing methods which are followed by the use of classifiers in most cases. Traditional image-processing algorithms use one or more image-processing methods to extract the important features of weeds and plants and then apply a classifier to those features to identify the specific weed and plant type (Fig. 4). Texture, shape, color, and spectral features of crops and weeds are the most common features extracted from the images [45].

Traditional Image processing methods include: a) Shape extraction; b) Texture extraction; c) Spectral feature extraction; and d) Color feature extraction.

(a) Shape extraction

Shape information in images can be investigated using shape descriptors. These are mathematical functions that include information

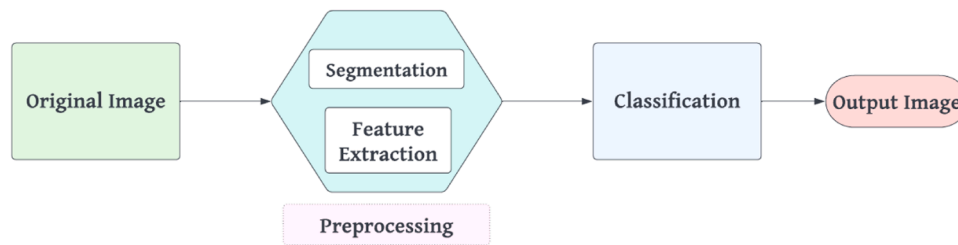


Fig. 4. General workflow of image processing-based weed detection.

about diameter, area, circularity, perimeter, eccentricity, rectangularity, etc. These descriptors can be broadly classified into contour-based descriptors and region-based descriptors [46]. Region-based shape descriptors include simple scalar region descriptors such as area, rectangularity, shape elongation, compactness, shape rectangularity, etc. [47]. Shape information could also be described using statistical properties such as moments. Hu's moment invariants are one of the most popular shape descriptors. It contains seven invariant moment parameters independent of scaling, rotation, or geometric translation [45].

dos Santos Ferreira et al. [48] used a method called simple linear iterative clustering (SLIC) superpixel segmentation to group pixels based on similarity in color and spatial proximity. They used this method to separate images into segments that contain multiple leaves of soy and target weeds. This is followed by classification using Convolutional Neural Networks (CNNs) which gave a high classification accuracy (99.5 %) as compared to the traditional ML-based classifier such as support vector machine (SVM). However, this research was performed in a controlled environment and the accuracy is bound to decrease in real-field applications with a more diversified dataset. Agrawal et al. [49] used 11 shape and five texture-based parameters followed by linear discriminant analysis to correctly classify weeds with an accuracy of 6–80 %. Herrera et al. [50] used a set of seven Hu moments along with six geometric shape descriptors (perimeter, diameter minor axis length, major axis length, eccentricity, and area) to classify weeds as monocots and dicots. This was followed by the use of four fuzzy logic-based decision-making methods for classification. However, the best accuracy achieved was less than 86 % and the classification only works in the early stage of weed growth when there is little overlap among the leaves of the weeds.

(a) Texture extraction

Texture feature extraction helps in outlining the morphological differences in the leaves and stems of plants and weeds by capturing the correlation between neighboring pixels [51]. The variation in the texture of leaves because of variations in surface roughness, vein structure, and sharpness of the edges of the leaves also helps us differentiate different weeds and plants based on the texture of leaves [52]. In addition to the difference in the texture of the leaves, the stem and flowers of the weeds have a different texture in the case of nutsedge, grass leaf, and broadleaf weeds.

There are multiple ways to perform texture-based classification and these methods can be broadly categorized into four categories: statistical methods, structural methods, model-based methods, and transform-based methods. Statistical methods include popular methods such as co-occurrence matrix methods (Gray Level Co-occurrence Matrix, known as GLCM) is the most common), Local Binary descriptors, and histogram properties. Energy variation and edge-based texture granularity detection are some examples of structural methods, the autoregressive model and fractal methods are some examples of the model-based method, and wavelet transform and curvelet transform are methods of the transform-based approach of texture classification method.

dos Santos Ferreira et al. [48] used GLCM to capture the texture of

the images of weeds in soybean plantations and used texture properties such as energy, contrast, correlation, and homogeneity in the GLCM matrix. They also used an LPB extractor on all pixels because of its invariance to gray-level fluctuations. Three classifiers (SVM, Adaboost – C4.5, and random forest) were used and their performance with each extractor was analyzed. SVM performed best with LBP while the other two performed best with color extractor. On this basis, they concluded that texture was an essential factor in correctly differentiating soybean and weeds. Bakhshipour et al. [53] used the Haar wavelet transform to obtain four sub-images with extractable features. This was followed by wavelet transform to extract 52 wavelet-texture features for texture analysis of weed patches. A detection rate of 96 % was obtained for weeds among sugar beet crops. But, since the detection was based only on the textural features, this would work only in cases where there are significant textural feature differences in weeds and plants. Major challenges associated with the texture-based methods are sensitivity to changes in noise, resolution, rotation, and issues with computational complexity.

(a) Spectral feature extraction

Spectral feature extraction relies on using the differences in spectral indices and reflectance values of different types of weeds and plants. Louargant et al. [54] captured multispectral images and used spectral mixing in the pixels to estimate the effect of image resolution on the ability to discriminate tiny weeds and plants. Hyperspectral imaging can be used to capture the near-infrared (NIR) wavelength from outdoor farmland objects such as crops, roads, trees, and soil and use this information to classify these objects based on spectrum analysis. A classification model was created using optimized reference pixels in the region of interest (ROI) acquired and it was followed by classification using maximum likelihood and support vector machine algorithms. The reported classification accuracy based on the reference pixels was 97.5 % [55]. This research was however limited to farmland objects and images with fuzzy boundaries would affect the performance badly.

(a) Color extraction

Burks et al. [56] used Hue, saturation, and intensity (HSI) color features to discriminate between six types of weeds and soil. Color co-occurrence matrix (CCM) method-based models were created, and discriminant analysis was used to determine the combinations of CCM features that gave the best classification accuracy. Following this, a discriminant classifier was trained using the previously generated models to conduct classification tests on each weed type. It was determined that the model that combined hue and saturation gave the best classification accuracy (93.3% overall), while the one with only intensity as the feature, gave the worst performance. Many shape-based plant species recognition studies use color segmentation as a first step to distinguish plants from the soil. Few have found high levels of success in using color segmentation alone to distinguish crops from weeds in ground-based field images [57]. Rasmussen et al. [58] employed a detection procedure based on color analysis that could differentiate green and senescent vegetation in UAV-acquired RGB images. They were

able to correctly classify 92–97 % of weed patches in wheat and barley fields under varying environmental conditions. However, the accuracy was significantly lower than 84 % in cases where there were other green crops and green vegetation. Some examples of previous research work that have employed various feature extraction methods for weed detection are presented in Table 2.

4.2.2.2. Detection using object detection algorithms and deep learning methods. The earlier approaches to weed and crop differentiation utilized feature extraction from image processing methods followed by an ML-based classifier to classify each image pixel. Even with successive improvements over the years that increased the performance of these algorithms, they still suffer from poor performance issues that affect the feature extraction methods because of poor lighting, occlusion, sensitivity to noise, and clustering. Deep learning (DL) is a subset of ML that differs from ML in the way it learns the information and quantity of data each algorithm uses [59]. Deep learning uses a backpropagation algorithm to discover intricate structures in large datasets and uses this information to determine how the machine should modify its learning, as well as better, compute the representation in each layer by learning from the previous layers [60].

Deep learning has had applications in many fields. It has been applied in health and medicine for imaging diagnosis, drug design, identification, and classification of cancerous and non-cancerous cells [61]. It has seen applications in many applications that involve computer vision [62] from object detection (in autonomous cars, robots, industrial robots, and UAVs), object tracking security systems [63], computers and cell phones for face detection, in facility management for monitoring heating and ventilation [64]. Deep learning has found wide

Table 2
Feature extraction methods with advantages and disadvantages.

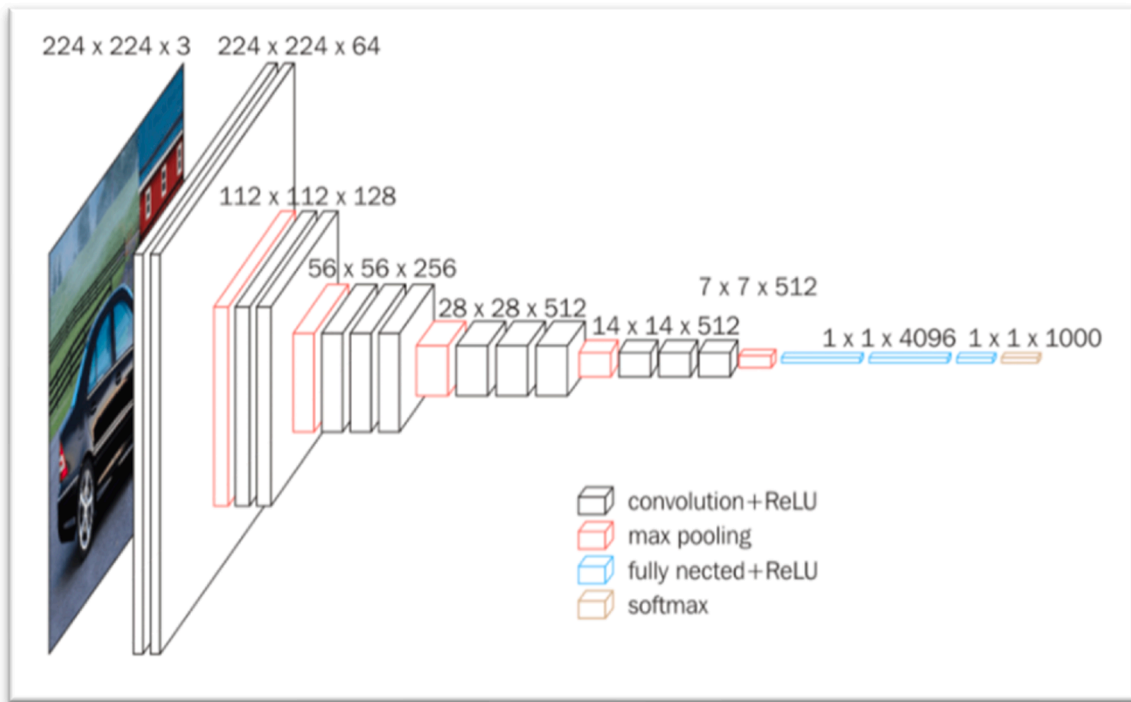
Feature extracted	Advantages	Disadvantages	Best performance (examples)	Refs.
Shape	Easy to implement, intuitive, unaffected by the time of capture and lighting, robust to noise and translation	Occlusion severely hinders efficiency, and similar shapes of leaves cause problems. Shapes deformed by disease and insects. Generally combined with other methods to improve performance.	99.5 % classification accuracy. (In combination with CNN)	[48]
Texture	Invariance to gray level fluctuation; GLCM is easy to implement; HOG is rotation invariant and has a low computational complexity	GLCM has a high processing requirement due to large dimensionality. It consumes a high amount of memory (Humeau-Heurtier, 2019).	96 % accuracy in the detection of weeds among sugar beet	[53]
Spectral	Provide sufficient information to distinguish crops from the soil. Good in partial occlusion	Fail in early stages when crops and weeds have similar reflectance. The time of capture also affects the values. Need for expensive spectral cameras	97.5 % classification accuracy	[55]
Color	Simple steps to extract when the color difference is obvious; Insensitive to changes in scale, size, and position	Affected by changes in lighting conditions, leaf disease, and similar colored crops and weeds	92–97 % accuracy in wheat and barley fields	[58]

adoption in the field of agriculture in fruit detection [65], disease detection in crops; [66–68], yield prediction and mapping [69,70], crop classification [71], fraud detection [72], and weed detection [73,74]. Deep learning methods make feature extraction redundant and can use semantic and spatial feature differences to improve weed identification and differentiation from crops [45]. In this section, we will review the convolutional neural networks (CNNs)-based DL methods and non-CNN-based DL methods applied in weed detection. The non-CNN methods include fully convolutional networks (FCNs), semi-supervised, and unsupervised feature learning.

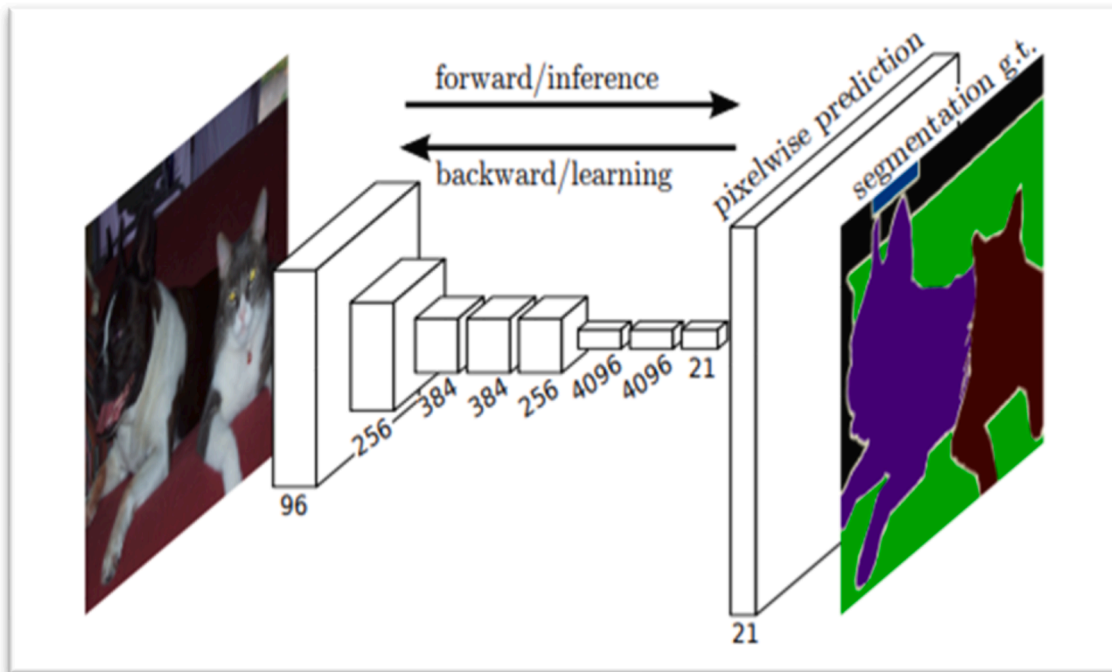
(a) CNN-based weed detection

The concept of neural networks was originally devised in the 1940s to mimic the human brain to solve general learning problems. Deep learning is a neural network with deep layers or structures that have become popular since 2006, after achieving breakthrough performance in speech recognition. The improvements in technology related to processing power, parallel computing, improvement in network structures and training strategies, and the availability of large-scale annotated data have also accelerated the popularity and widespread adoption of deep learning within the scientific community. CNNs are the most typical form of deep learning [75]. CNNs are multilayer neural networks comprised of different types of neural layers such as convolutional layers, pooling layers, dropout layers, rectified linear unit layers (ReLU), and fully connected layers (Fig. 5(a)). These layers work together to extract features from the input image, reduce the dimensionality of the feature map to keep only the important features, and perform classification based on the features extracted in the previous layers [76].

There has been a significant rise in the use of CNNs for weed classification and detection in the past decade. Burks et al. [78] compared three neural network models-backpropagation, counter propagation, and radial basis function to determine the model with the best classification accuracy for weed species classification. It was established that the backpropagation neural network classifier achieved the highest classification accuracy (96.7 %) as compared to the other neural network methodologies. Dyrmann et al. [79] built a new architecture involving three convolutional layers, two maximum pooling layers, two residual blocks, and two fully connected layers. This network achieved a classification accuracy of 86.2 % when tested on over 10,000 images containing two weeds and crop species at the early growth stage. Potena et al. [80] utilized two different CCNs (named *sNet* and *cNet*) for fast and accurate weed and crop classification on RGB and NIR images. *sNet*, the lightweight CNN had a single convolutional layer followed by a rectified linear unit (ReLU) activation function and maximum pooling layer which helped with vegetable segmentation. The *cNet* was the deeper CNN and was used for pixel-wise classification of weeds and crops. Beeharry and Bassoo [81] investigated the performance of artificial neural networks (ANNs) and AlexNet on UAV-based images to classify soil, soybean, grass, and broadleaf weeds. AlexNet algorithm gave a classification accuracy of 99.8 % as compared to the ANN algorithm which gave an accuracy of under 50 %. This high-level performance of AlexNet was attributed to it being a deep learning-based CNN algorithm with enough layers and computational complexity to achieve high efficiency in weed detection. To improve the processing speed by reducing the number of parameters, McCool et al. [82] used three stages in their approach. They started by employing a pre-trained model followed by using a teacher-student network to compress models and learn a less accurate lightweight CNN which had fewer parameters. Finally, multiple lightweight models were combined to improve performance and achieved an accuracy greater than 90 % along with a processing speed of 10.7–1.83 frames per second. You et al. [83] proposed a deep neural network with a hybrid dilated convolutional network, a DropBlock to learn robustness features, and two additional blocks to capture the global discriminative features. The network was tested on two datasets – Stuttgart and Bonn and achieved a mean intersection over union (mIoU)



a)



b)

Fig. 5. (a) CNN architecture (VGG16) (<https://neurohive.io/en/popular-networks/vgg16/>), (b) FCN architecture [77].

of 88–89 % on both datasets. Milioto et al. [84] provided four vegetation indices and other representations of the raw input such as HSV color space, Laplacian, Canny edge detector, etc. as inputs to the raw RGB image. This augmented input was then passed to the CNN which performed a better pixel-wise segmentation, generalized to newer datasets well, and improved the real-time speed (20 Hz). It achieved a high precision - 98.16 % for weeds, and 95.17 % for crops on the Bonn

dataset. Graph Convolution Networks (GCN) is a variation of CNNs that operate on the graph data structure [85]. GCN combined with a Resnet 101-based feature extractor was used by Jiang et al. [86] to improve weed detection when the dataset is limited. They constructed a GCN graph based on semi-supervised learning which utilized the extracted weed CNN features and their Euclidean distances. The proposed approach achieved a high accuracy of 96–99 % on four different weed

types and outperformed the state-of-the-art methods (AlexNet, VGG16, and ResNet-101). Bah et al. [87] used an unsupervised training dataset with CNN for weed detection in bean and spinach fields. The results were similar to the supervised data labeling with the area under the curve (AUC) difference being 1.5 % and 6 % in the spinach and bean field respectively.

The above CNN-based classification and detection methods perform better than the ones that rely entirely on image processing and manually extracted features and can work well with variations in environmental conditions during image capture and real-time testing.

(a) Non-CNN-based weed detection

Although CNN is the most popular algorithm for weed detection and object detection in general, other algorithms such as FCNs, Region Proposal Networks (RPN), ANNs, and semi-supervised and supervised feature learning also have potential. FCNs (Fig. 5(b)) do not have dense layers and fully connected layers that are generally found in a CNN. The absence of a fully connected layer means any size of input image could be used [77]. Bakhshipour and Jafari [88] implemented two popular algorithms—SVM and ANN for the autonomous detection of weeds in a sugar beet field. ANN performed better in crop weed discrimination (R^2 of 93 % on train data, 92 % on test data) as compared to plant type identification (R^2 of 84 % on train and test data). Both the classifiers had a low percentage of false negatives which is quite promising for plant detection among weeds to avoid spraying on plants. Dyrmann et al. [89] used a FCN trained on images with more than 17,000 annotations to detect weeds in color images with heavy occlusion. Ma et al. [90] proposed a robust image segmentation method using SegNet which is fully based on FCN, for semantic segmentation of rice seedlings and weeds in the paddy fields. The SegNet-based method was compared to the classical semantic segmentation methods such as FCN and U-Net models and achieved a high accuracy rate of 92.7 % as compared to the latter methods which achieved an accuracy of 89.5 % and 70.8 %. CNNs and FCNs are supervised learning algorithms that need annotated data from thousands of images. Since in many cases, it is not possible to collect large datasets of each of the classes involved, many researchers tried using semi-supervised and unsupervised learning to achieve weed detection. Hu et al. [91] developed a novel semi-supervised deep architecture called Graph Weeds Net (GWN) which can formulate the weed patterns from multiple graphs and calculate relations among graph vertices. This approach only needed image-level annotation (images with boundaries drawn on target areas) and not patch-level annotation (refers to image patches with a class label). Evaluation of the network on the DeepWeeds dataset resulted in an accuracy of 98.1 %. Some examples of previous research work that have employed DL methods for weed detection are presented in Table 3.

(a) General issues with the DL methods

Table 3
Some examples of high-performing deep learning-based methods.

Author (s)	DL type	Performance	Network features
[81]	CNN-based	99.8 % classification accuracy	AlexNet with encoder and decoder
[80]	CNN-based	mAP of 98.7 %	lightweight CNN – sNet. Deep CNN cNet for classification
[91]	Semi-supervised	98.1 % accuracy on the DeepWeed dataset	Semi-supervised Graph Weed Network
[86]	Semi-supervised	96–99 % accuracy	Semi-supervised GCN
[82]	CNN-based	More than 90 % accuracy and an FPS between 1.07–1.83	Resnet-101 pre-trained Deep CNN with 25 M parameters

The biggest issue with DL methods is the requirement of a large dataset and the subsequent time and effort involved in labeling the dataset. Researchers have tried to tackle the issue of labeling efforts through unsupervised learning by utilizing the spatial distribution of crop rows to obtain a training dataset that can distinguish crops from weeds based on spectral data. This training dataset allowed automated spectral classification on the crop row pixels to improve the weed detection rate [92]. However, this works under the assumption that most weeds are outside the rows which is not always true. In cases where the available data is limited, previously trained models could be used through transfer learning. A detection database with sufficient variability in terms of images captured from different cameras with different resolutions helps in achieving improved detection networks that deliver better performance [93]. The dataset should also capture different growth stages under different lighting conditions, shadows, and noise. When there is a shortage of training data with enough variability and a balance of required classes, data augmentation techniques are necessary. This improves the learning procedure, the generalization, and the overall performance of the model as it is fed a varied and balanced dataset [94].

4.2.2.3. Spraying system. The spraying system is the other major subsystem of the precision smart sprayer or the weed sprayer system. The spraying system receives the information from the machine vision and control system and activates specific nozzles from an array of nozzles that target specific weeds while avoiding the crop. A flowchart of the components of a typical spraying system is shown in Fig. 6.

4.2.2.4. General setup of the spraying system. Lee et al. [41] developed a robotic spraying system for tomato plants with eight 12 V DC solenoids, a specially designed accumulator (to maintain a constant flow rate), and driver circuits for valve control. Instead of using custom nozzles, five 22-gage stainless steel hypodermic tubes (0.28 mm I.D.) were used to form a micro-spraying nozzle. Outdoor tests were conducted with real tomato plants and a green circular patch representing the weeds, while indoor tests were conducted using rectangular plant targets and circular green targets. The system had poor weed spraying accuracy (52.4 %) mainly because of incorrect identification and avoidance of spraying on weeds near plants. However, they had good spraying accuracy on the circular green targets where the average error between the center of the target and the spray drops was 6.58 mm.

Lamm et al. [96] developed an automated weed control technology using the robotic platform from Lee et al. [41] to detect the cotton crop and weed locations. The spray nozzles were located 35 cm behind the camera. Field trials were conducted in four different locations using a mixture of water and blue dye applied while moving at a speed of 0.45 m/s. The system correctly sprayed 88.8 % of the weeds while avoiding 78.7 % of the cotton plants.

The precision sprayer developed by Tian [97] used a 7.3 m wide boom section with 12 solenoid valves mounted 50.8 cm apart mounted on a Patriot XL sprayer (CASE-Tyler Industries Inc., Benson, MN). The sprayer was self-propelled and map-driven, with the boom pressure set to 172 kPa to achieve a flow rate of 3.8 liters/min. Under three different ranges of travel speeds between 3.2 – 17.7 km/h (3.2 – 3.9 km/h, 6.9 – 8.7 km/h, and 11 – 14 km/h), a target hit accuracy between 86 and 96 % was achieved, with the best performance coming in when the sprayer speed was between 3.2 km/h to 3.9 km/h. However, the spray pattern lengths were high for the targets, i.e., the distance of the start and end of spray in the fore and aft direction from the center of the target was high and followed a distribution with a mean of 0.543 m and 0.227 m respectively.

Søgaard and Lund [98] developed an autonomous robot with a micro-dosing system for spraying. The micro-dosing system consisted of a micro-boom (100 mm wide) with 20 evenly spaced tubing (250 mm long, 0.5 mm I.D., polytetrafluoroethylene) situated 50 mm above the

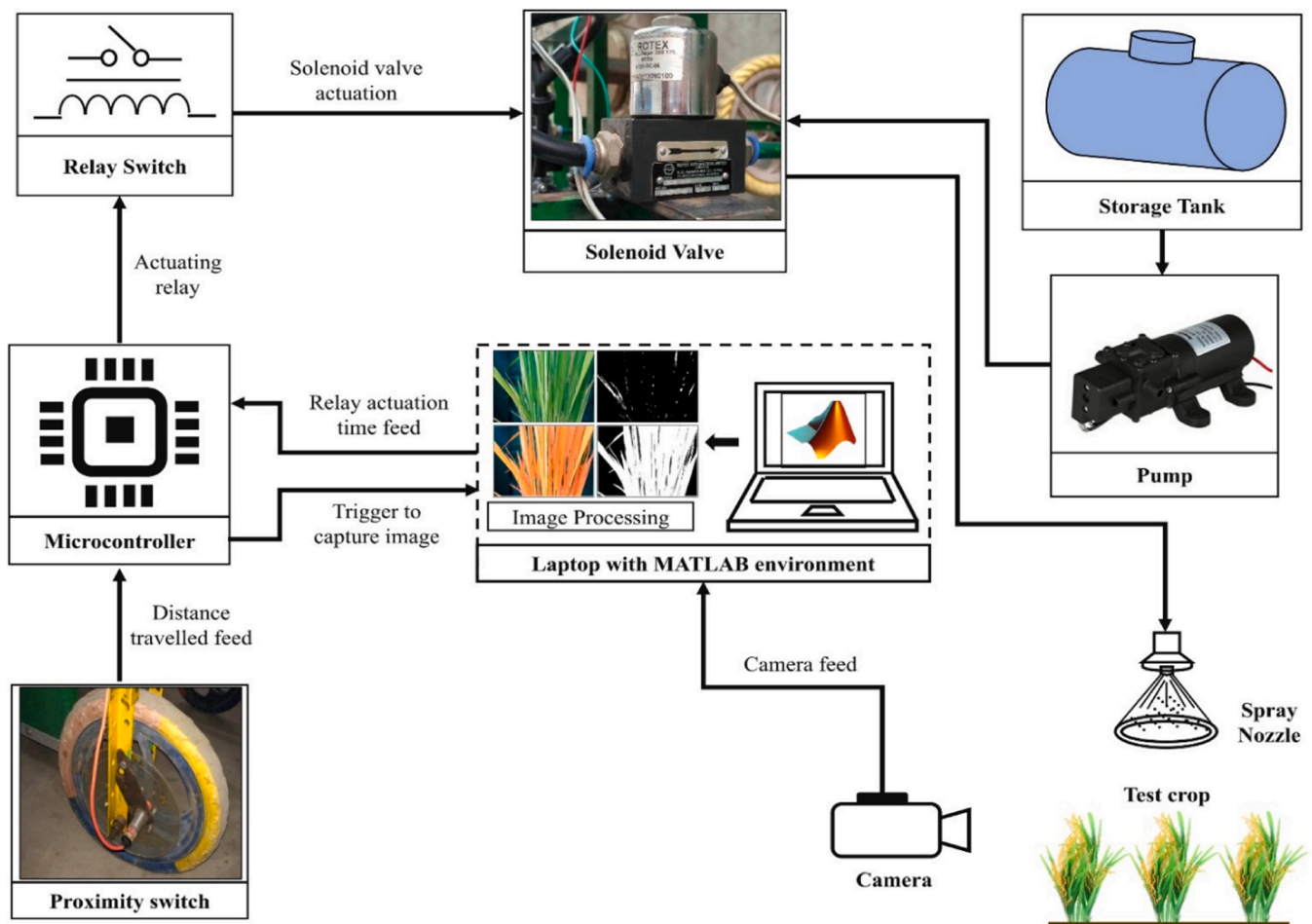


Fig. 6. Flowchart of the spraying system components [95].

floor. 12 V DC solenoid valves were used to open and close the valves. Experiments were performed under controlled indoor environments where black circular disks of various sizes were used on PVC sheets to test the spraying accuracy and resolution while moving with a forward speed of 0.2 m/s. The spraying system achieved a spatial spraying resolution of 5 mm by 5 mm and 99.8 % of circles were hit by at least one shot. Although these results are encouraging, there were many off-target shots (27 %) that could harm the plants in a real-life field application.

Machine vision-based detection and spraying systems have also been employed in sugar beet fields for the control of volunteer potato fields [99]. They used five needles each spaced 40 mm apart and at a height of 0.3 m above the soil level. Based on the detection from the machine vision system, pulse width modulation (PWM) signals were sent to fast-acting solenoid valves to open and close the needles to spray 20 ± 5 μ l droplets of 5 % (v/v) glyphosate solution. Laboratory tests in a controlled environment resulted in an accuracy of 14 mm in the longitudinal direction and 7.5 mm in the transverse direction. For field tests at 0.2–0.8 m/s, the accuracy in successfully avoiding sugar beets was high (99.2 % on sandy soil and 99.6 % on clay soil) while the hit rate of volunteer potato plant was comparatively lower at (88.3 % on sandy soil and 85.7 % on clay soil). However, the spray resolution was low at 10×40 mm.

Rehman et al. [100] used flat fan TeeJet nozzles operating under a pressure of 275 ± 10 kPa on their variable rate sprayer (VRT) for real-time field evaluations. The nozzles were mounted on a sprayer boom 1.22 m above the ground (Fig. 7(b)) and a flow rate control mechanism using a Land Manager-II controller, a Garmin PC receiver, a flow meter, and servo valves-controlled flow to the nozzles based on

speed information. Field evaluations were focused on the spray area on the goldenrod patches as compared to the total area to determine the potential savings in chemical application. Actual savings were reported to be between 30 and 60 %. However, there was no evaluation of the actual target hit rate, missed targets, and the effect of speed on accuracy.

Partel et al. [74] developed a low-cost smart sprayer using three webcams and 12 TeeJet nozzles to cover a work length of 1.08 m. Each nozzle was controlled by 12 V solenoid valves which had a response time of less than 50 ms. An RTK GPS was used to determine the target position for spraying purposes. Experiments were performed using two different GPUs (selected based on processing power and cost) with artificial plants and weeds and real plants and weeds in the field. Using the GTX 1070 Ti GPU, 91 % of targets were fully sprayed and no non-targets were sprayed in the case of artificial plants. However, in the case of real plants, the non-target spraying percentage was 14–15 % for both GPUs. The system also suffered from higher missed spraying targets to the right of the sprayer. Table 4 presents, in a tabular form, the sprayer-specific data from the previous research discussed in this section.

4.2.2.5. Fluid dynamics and controls of precision sprayers. Previous research in precision weed sprayers has focused more on the machine vision aspect without delving deep into the fluid dynamics associated with sprayers or on the occasions when the fluids aspect was considered for the spraying systems. The focus was more on the controls of the nozzles and fluid dynamic analysis of spray droplets. Paice et al. [102] investigated different sprayer dose control systems and discussed the constraints of each system that make its implementation difficult. Accuracy, speed of response, and turn-down characteristics were used as



Fig. 7. (a) Micro-sprayer system with 5 needles [99]; (b) Variable rate sprayer mounted on a John Deere gator [100]; (c) VR sprayer [101]; (d) Ecorobotix sprayer.

the key performance parameters. On-off solenoid valves provide high lateral resolution and have rapid response times. For a typical direct-acting solenoid valve, the pressure drop across the orifice (ΔP_o), the force on the plug induced by hydraulic pressure (F_p), and the force on the plug by a magnetic field (F_B) are given by:

$$\Delta P_o = \frac{\rho}{2} * \left(\frac{Q}{C_o * A_o} \right)^2 \quad (1)$$

$$F_p = \Delta P_c * A_o \quad (2)$$

$$F_B = \frac{N I A_c}{2 D_g} \quad (3)$$

where

ρ = density of the fluid.

Q = flow rate

C_o = orifice loss coefficient

A_o = area of the orifice

A_c = magnetic core's cross-sectional area.

ΔP_c = valve opening pressure

N = number of turns on solenoid coils

D_g = gap in the magnetic circuit when the valves are de-energized

I = current through the solenoid

Paice et al. [102] also believed that using a pulse width modulation (PWM) control on solenoid valves could reduce the power losses in the

regulator element as well as the power on/power off time ratio during the valve hold-on period. The issues with pressure control systems, which suffer from reduced target deposits and non-uniform target doses, are also put forward. Han et al. [103] evaluated the volumetric application accuracy of a PWM-controlled variable-rate sprayer that had 25 nozzles mounted on an 18.3 m long sprayer boom. The variation in flow rate was measured for each nozzle at four different duty cycles (25, 50, 75, and 100 %) with a set system pressure of 310 kPa. Based on the data obtained, the average flow rate curve was obtained and the relative flow rate control error (defined in Eq. (4)) was used for the overall evaluation of the system.

$$E_r = 100 * \frac{Q_m - Q_a}{Q_m} \quad (4)$$

where

Q_m = measured flow rate

Q_a = actual flow rate

E_r = relative flow rate error.

The primary reason for a high error in relative flow rate control was identified to be the valve-to-valve differences in flow rate responses to PWM signals. Aiwu et al. [104] designed an on-line flow control system for a VRT system using a proportional integral differential (PID) control algorithm. A variable rate flow control system was developed using a National Instruments (NI) data acquisition system (DAQ), a flow meter, and an electronic control injector of a vehicle. Based on the experimental measurements it was determined that the flow rate (Q) of the injector

Table 4

Examples of spraying systems with their main components, features, and main advantages/disadvantages.

Author (s)	Crop (s)	Main components	Control type	Speed of travel	Nozzle height (above beds/ground)	Spraying time and accuracy	Advantages/disadvantages
[41]	Tomato	Custom-built Capstan spray valves, eight 12 V DC solenoids, accumulator, CO ₂ tank to pressurize the system (103 kPa)	Sensor Watch, TERN, Inc. with encoder and computer input	0.8 km/h (outdoor test)	101.6 mm	Avg error: 6.58 mm; spray time 10 ms; weed spraying accuracy 52.4 %	Poor weed spraying accuracy (52.4 %) due to detection constraints, low spraying error
[96]	Cotton	Custom-built Capstan spray valves, eight 12 V DC solenoids, accumulator, CO ₂ tank to pressurize the system (103 kPa)	Sensor Watch, TERN, Inc. with encoder and computer input	1.62 km/h	101.6 mm	88.8 % weed spraying accuracy	Improved weed spraying accuracy. Crop spraying accuracy is still low (21 % sprayed).
[97]	Corn and soybean	Synchro 12 V solenoid valves; ground speed radar gun, TeeJet 8006VS nozzles, boom pressure (172 kPa)	V-25-based (NEC Electronics Inc., Mountain View, CA) embedded microcontroller	3.2 - 3.9, 6.9 - 8.7, 11 - 14 km/h	355.6 - 381 mm	86 - 96 % hit accuracy	High hit rate. A very large spread of spray (269.2 mm and 261.6 mm) from the center of the target to the left and right edges of the spray
[98]	Indoor experiments	20 evenly spaced tubing controlled by 12 V DC solenoid valves	A computer that controlled the robot also controlled the spraying system	0.72 km/h	50 mm above the floor	27 % off target shots	Sub-centimeter accuracy, 99.8 % of circles hit by at least one shot. Greater missed targets.
[99]	Sugar beets with volunteer potato plants	5 micro sprayer needles controlled by DC solenoids	PWM signals from the real-time operating system	0.72 - 2.88 km/h	0.3 m	High avoidance of plants (>99 %)	Comparatively lower target hit rate and low spray resolution
[100]	Blueberry	Eight TeeJet nozzles, Dickey John LMC, Garmin receiver, flow meter	U3-HV I/O module sent triggering signals	No information	1.22 m	Savings on herbicide between 30.12% –60.58 % based on spray area	Spraying accuracy and hit rate evaluation missing
[74]	Pepper	12 TeeJet nozzles, 5 V relays, RTK GPS	Arduino controlled based on Jetson signals	Variable vehicle speed	Not mentioned	High precision (95 %) and recall (89 %) of the spraying system	Non-targets were sprayed; missed targets were concentrated to the right of the sprayer

increased linearly with the square root of the feedback voltage (V) as given by Eq. (5).

$$Q = 0.4092 \sqrt{V} + 0.2046 \quad (5)$$

Researchers have also measured the initial spray characteristics of agricultural nozzles [105]. The theoretical nozzle discharge velocity was estimated using the nozzle pressure (P_n) and density of spray liquid (ρ_L) (Eq. (6)) and it was experimentally determined using an Oxford Lasers imaging system (Oxford, UK), and particle image velocimetry software (VidPIV 4.0).

$$U_L = \left(\frac{2P_n}{\rho_L} \right)^{0.5} \quad (6)$$

The theoretical and experimental results for discharge velocity at nozzle pressures of 150, 280, and 450 kPa compared well with each other. Initial droplet size, spray density, and fan angle were also measured. Although air induction nozzles (TeeJet nozzles; TeeJet, Denmark) had the narrowest fan angle, all the nozzles had an increasing fan angle with an increase in pressure. The XR11002 (TeeJet Extended range flat-fan; TeeJet, Denmark) produced the smallest droplet size and the highest velocity. The most common method used to characterize the droplet size distribution in terms of variables is the Rosin-Rammler distribution equation (Eq. (7)) [106].

$$1 - \nu = \exp \left[- \left(\frac{D}{\delta} \right)^\gamma \right] \quad (7)$$

where

ν = fraction of the total volume contained in droplets smaller than diameter D
 D = droplet diameter

δ = Rosin-Rammler diameter

γ = a constant representing the amount of spread of droplet sizes

[106] also suggested that the Lagrangian model is the most common approach to simulate the behavior of air-assisted pesticide sprayers and estimate the movement of the droplets in both laminar and turbulent flow fields. They believe in the need for developing simple models for open fields as opposed to controlled environments where the assumptions of uniform droplet sizes and uniform velocities would not hold. Duga et al. [107] used a 3D computational fluid dynamics (CFD) model of spray drift from orchard sprayers to determine airborne drift considering tree architecture, tree canopy wind flow, and sprayer movement. Unsteady Reynolds averaged Navier-Stokes (URANS) equation and k - ϵ turbulence model were used to model wind and airflow from the sprayer. A 50 % reduction in spray deposition close to the trees was achieved when Albux ATR orange hollow cone nozzles and Albuz TVI 8002 drift-reducing nozzles were used together in the CFD simulation. Mangus et al. [108] evaluated a commercial PWM system for individual nozzles for a 6.6 m long three-section boom sprayer with 13 nozzles. The static spray tests were compared with dynamic simulation results obtained from a simulation software developed using Microsoft Excel and MATLAB and a test protocol was created to characterize the nozzle flow and deposition. It was determined that duty cycles over 60 % would provide better spray coverage uniformity for PWM systems. However, the experimental spray tests were conducted in laboratory conditions, and field studies are needed to validate the dynamic simulation results for real-time application.

5. Discussion and future scope

Precision weed management utilizing smart sprayers is an efficient approach for reducing the overuse of herbicides, reducing herbicide-

related environmental pollution, and working towards a sustainable food production system with more systematic weed control and an increase in crop productivity.

5.1. Challenges of non-herbicide-based weed control systems

Although non-herbicide-based weed control systems may be safer for the environment, they have their unique challenges and limitations that prevent their widespread adoption in the field of weed control. The majority of the older mechanical weeders did not have a mechanism to differentiate between the weeds and crops and hence only worked in between plant rows. Most of the mechanical weeders that employ harrows require multiple passes to be effective and work only on weed growth between rows. Utilizing them on raised beds covered with plastic will tear off the plastic and promote more weed growth on the beds. Mechanical weeders that use harrows and finger weeders need softer soil conditions for them to penetrate the soil and uproot the weeds while the weeders with punching systems that push the weeds down into the soil would not work well on stony and lumpy soils.

Laser treatment to kill weeds requires it to be applied at an early stage to be effective. Smaller powered lasers require more time to effectively cut or burn the weeds while the lasers with more power capacity could damage the nearby plants due to laser beam splitting. Also, it could damage the plastic covers if used on crops grown on raised beds with plastic covers. To control larger weeds, high-power lasers are required, and they must provide sufficient energy in short durations to be relevant in large-scale production [109]. Thermal weeding systems with hot water and steam require additional tanks to store hot water, a heating mechanism, and the power required to keep the heating systems going for longer application times. They also have an added safety risk because of the high temperatures involved.

5.2. Challenges of the machine vision aspect of precision sprayers

Researchers have been focusing efforts on weed identification and management through non-chemical and chemical use for the past three decades. With the advances in machine learning and deep learning algorithms, the increased processing power of computers, low-cost graphics processors, cheaper high-resolution cameras, and spraying components, the field of precision spraying has seen a lot of developments in the past decade. There has been a shift from using standard image processing methods to extract features (followed by using a robust machine learning algorithm as a classifier), to using more advanced deep learning algorithms that have in-built feature extractors. Consequently, the detection accuracy and processing speeds have increased. However, even the top DL algorithms also have their disadvantages in terms of the requirement of large datasets for training, the cost associated with data annotation, higher memory requirements, and slower training speeds. The use of generative adversarial networks (GAN) [110] and synthetic data generation along with the availability of many public datasets have reduced the need for collecting a large amount of data to some extent. The use of semi-supervised and unsupervised algorithms [111] also has tried to negate these disadvantages, but their widespread adoption has been restricted as most of the research is still focused on deep learning and not on the capability of deep neural networks to transfer knowledge in a supervised and semi-supervised learning [112].

Dataset collection also presents a few challenges related to data quality. To achieve higher accuracy, the dataset needs to have a high percentage of data that closely resembles the actual test conditions expected in the field. However, to improve the generalizability of the model, the dataset should include images collected from different locations with different lighting conditions, backgrounds, noise levels, weed and plant growth stages, shadow conditions, and in-field conditions (soil types, wind levels, etc.) [113]. This can be achieved by either capturing the images in those conditions or through data augmentation

techniques. The availability of open datasets for plants and weeds helps to some extent, but they might not capture the varying conditions found in the field.

5.3. Challenges of the spraying systems of precision sprayers

As compared to the weed detection and machine vision aspect of weed sprayers, there has been less research done into the fluids, controls, and actual field-testing aspects of ground-based weed sprayers. Researchers have used CFD simulations to model the drift and ground deposition in a canopy sprayer (e.g., [107]) but there haven't been a lot of modeling approaches adopted to improve the efficiency of weed sprayers for crops and vegetables. Although there have been studies that evaluate the spraying characteristics in the field using colored papers or water-sensitive papers (e.g., [114]), most of the testing of spraying systems is done in controlled laboratory environments and not in the field where environmental conditions could affect the spray patterns and efficiency. Many researchers have used colored papers and circular disks to simulate weed presence and to evaluate spraying accuracy and resolution. While the use of colored papers and discs makes it easier to measure the spraying accuracy and resolution in a controlled environment, it is not a proper representation of the variations in spraying when affected by conditions such as wind, disturbances due to uneven ground, and vehicle speed variations. Reduction in detection accuracy of the weeds in the field due to varying light conditions, plant occlusion, and damages to plants and weeds due to weathering and pests is also bound to affect the spraying accuracy. Lower spraying accuracy and higher target miss rate are common issues faced in the previous spraying systems. Also, avoiding the spray drifting onto the plants when the weeds are close to the plants is still a major challenge especially when testing in field conditions.

5.4. Future scope

Over the past two decades, weeding systems have shifted from simpler inter-row mechanical weeders to more complex versions that employ machine vision-equipped weeding mechanisms capable of identifying and distinguishing crops and weeds. The detection accuracy has improved over the years as the industry has shifted from image processing and manual feature extraction methods to ML and DL algorithms. However, the issues associated with training the DL algorithms are still a major hurdle as it requires a large, annotated dataset. To negate the challenges of collecting large datasets of images for training the ML and DL models, the research community has made many datasets

Table 5

Some examples of publicly available datasets of plants and weeds.

Dataset	Crops/ plants/ weeds	Number of images	Reference
Quantitative Plant	Rice root, Apple leaf, Maize, Passiflora, etc.	1819,970	https://www.quantitative-plant.org/dataset
Weed25	25 different weed species	14,035	https://pan.baidu.com/share/init?surl=rnUoDm7IxxmX1n1LmtXNXw
Sugar Beets [115]	Sugarbeets	12,715	https://www.ipb.uni-bonn.de/data/sugarbeets2016/
DeepWeeds [116]	Eight different weed types	17,509	https://vision.eng.au.dk/
Open Plant Phenotype Database (OPPD)	64,292 individual plants from 47 different species	7590	https://vision.eng.au.dk/open-plant-phenotyping-database/

publicly available to the community. Table 5 gives a list of some of the datasets. Although the datasets cover a large variety of species of plants and weeds, we still need more datasets of images taken under different environmental conditions and bed backgrounds for multiple crops to make the training models more robust, more generalized, and less susceptible to noise. This would improve the transfer learning process rendering it more efficient and smoother. Processing power requirements and the associated costs also limit the development of the sprayers. Cheaper electronics and GPUs with higher processing power and storage capabilities are needed to make the systems more affordable to farmers. There are reasonably priced (less than \$2000) GPUs with high-performance processors from NVIDIA Jetson (Nano, Xavier, Orin) available for deep-learning applications, and they have been used by researchers for agricultural applications. But these are developer kits meant for initial prototyping and use in a controlled environment. NVIDIA doesn't recommend using them in extreme agricultural environments. There is a need for reasonably priced industrial-quality GPUs and processors that can withstand heat and dust conditions better than the existing developer kits.

More exploration needs to be done to improve fluid modeling of the spraying systems to prevent drift. Out of the existing flat fan nozzle types, the extended range flat-fan nozzle provides better spray distribution as compared to other flat-fan nozzles, however, their efficacy at different heights, wind speeds, and pressures needs further exploration in field tests. Improved nozzles with lesser cone angles and better control system design could reduce the drift of spray on the plants when weeds are too close to the plants. There is also a need for a comprehensive evaluation of the weeding robots in varying wind conditions, vehicle speeds, and soil conditions. Future work must include thorough field evaluations that consider real-world environmental factors which can provide valuable insights into the practical performance and limitations of autonomous weeding systems. There is also a need for delving deep into the significance of nozzle placements, the number of nozzles, the duration of spray, and its impact on spray patterns at different speeds of the machine in the field. Understanding how these precision spraying systems perform under diverse conditions will aid in their refinement and optimize their efficacy in different agricultural settings.

6. Conclusion

This study reviews the different methods used in precision weed management, machine vision techniques, and weed spraying systems. A total of 116 relevant papers and articles were examined based on their importance covering weeds in crop production, weed control without herbicides, weed control with herbicides, conventional weed management, and precision weed management subsystems, particularly machine vision and spraying systems. It was found that most earlier studies used image processing techniques based on either RGB or a combination of RGB with spectral cameras and performed feature extraction using multiple methods. This feature extraction was followed using a classifier. Maximum likelihood, random forest, and SVM were the most common ML algorithms used as classifiers. The application of DL algorithms in weed detection greatly increased in the past decade. This is because of the better performance of the DL algorithms as compared to the traditional image processing methods. Many DL-based classifiers and object detectors have more than 95 % accuracy. Spraying systems have been investigated in the past decades as well, but the amount of work done in this field for weed sprayers is less as compared to the weed detection and machine vision part of precision weed management. Many of the weed sprayers developed by the researchers were tested in controlled lab environments with color-based objects used as a substitute for weeds and sometimes plants. Research conducted on studying the fluid mechanics of the sprayers to improve the spraying efficiency is also limited. More work needs to be done to conduct actual field trials on raised beds and crop rows with the weed management systems to better gauge the adaptability of these systems to real-life conditions.

Ethical statement

The authors declare that this study does not involve human or animal subjects.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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